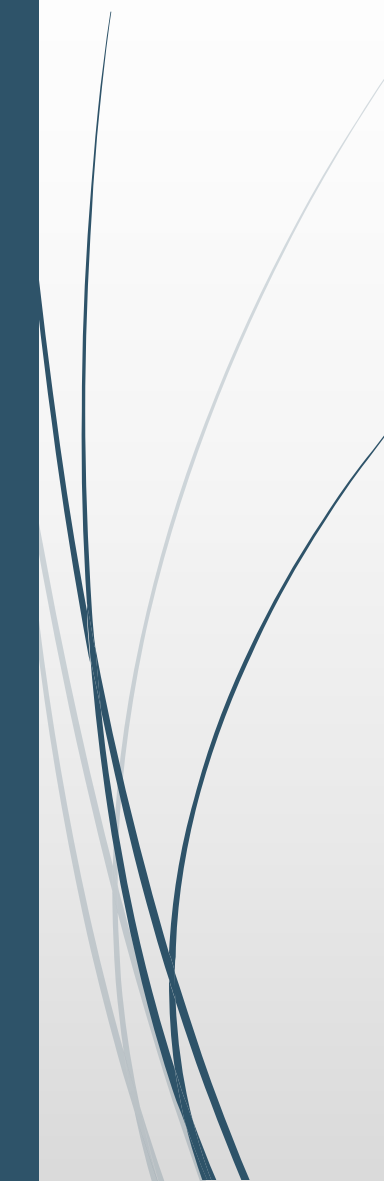




Overview



- Abstract
- Problem Statement
- Introduction
- Environmental Impact
- Literature survey
- Design
- Methodology
- Demo
- Conclusion
- References

Abstract

This project focuses on predicting the toxicity of chemical compounds using a hybrid approach that combines quantum machine learning and classical machine learning. Chemical data is represented using SMILES and graph-based structures to capture both molecular features and relationships. These features are encoded into quantum states and processed using quantum circuits with gates such as the Hadamard gate to generate intermediate outputs. The results are then passed to a classical machine learning model for training, testing, and final toxicity classification. The system aims to improve prediction accuracy and generalization for unseen chemical compounds by leveraging both quantum and classical techniques.



Problem Statement

Problem:

Researchers and pharmaceutical developers struggle to accurately predict the toxicity of new chemical compounds using traditional machine learning models, especially for unseen molecular structures.

Research gap:

This project proposes a hybrid approach using SMILES and graph representations combined with quantum machine learning and classical models to improve toxicity prediction and generalization.



Introduction

The prediction of chemical toxicity is an important task in drug discovery and environmental safety.

Traditional machine learning models using molecular descriptors and fingerprints derived from SMILES data are widely used for this purpose.

These models often struggle to accurately predict toxicity for unseen chemical compounds due to the complex and non-linear nature of molecular structures.

Recent advancements in quantum computing have introduced quantum machine learning techniques, which offer new ways to represent and process complex data, but their application in real-world chemical toxicity prediction is still in the early stages.

Literature survey

Author's Name	Paper Title	Chemical data	Method/model used	Key Findings
Suzuki et al.(2020)	Predicting Toxicity by Quantum Machine Learning	Phenol-based chemical compounds dataset	Quantum machine learning with quantum feature maps	Demonstrate that QML models can predict toxicity with improved features representation compared to classical models
W.-Y. Chiang et al. (2025)	Quantum Machine Learning for QSAR Prediction with Incomplete Data	Molecular descriptors from chemical compounds (QSAR datasets) with missing or incomplete features	Variation Quantum Classifiers (VQC) with quantum feature encoding	The study shows that quantum machine learning models can handle incomplete chemical datasets effectively and achieve competitive performance compared to classical QSAR models.

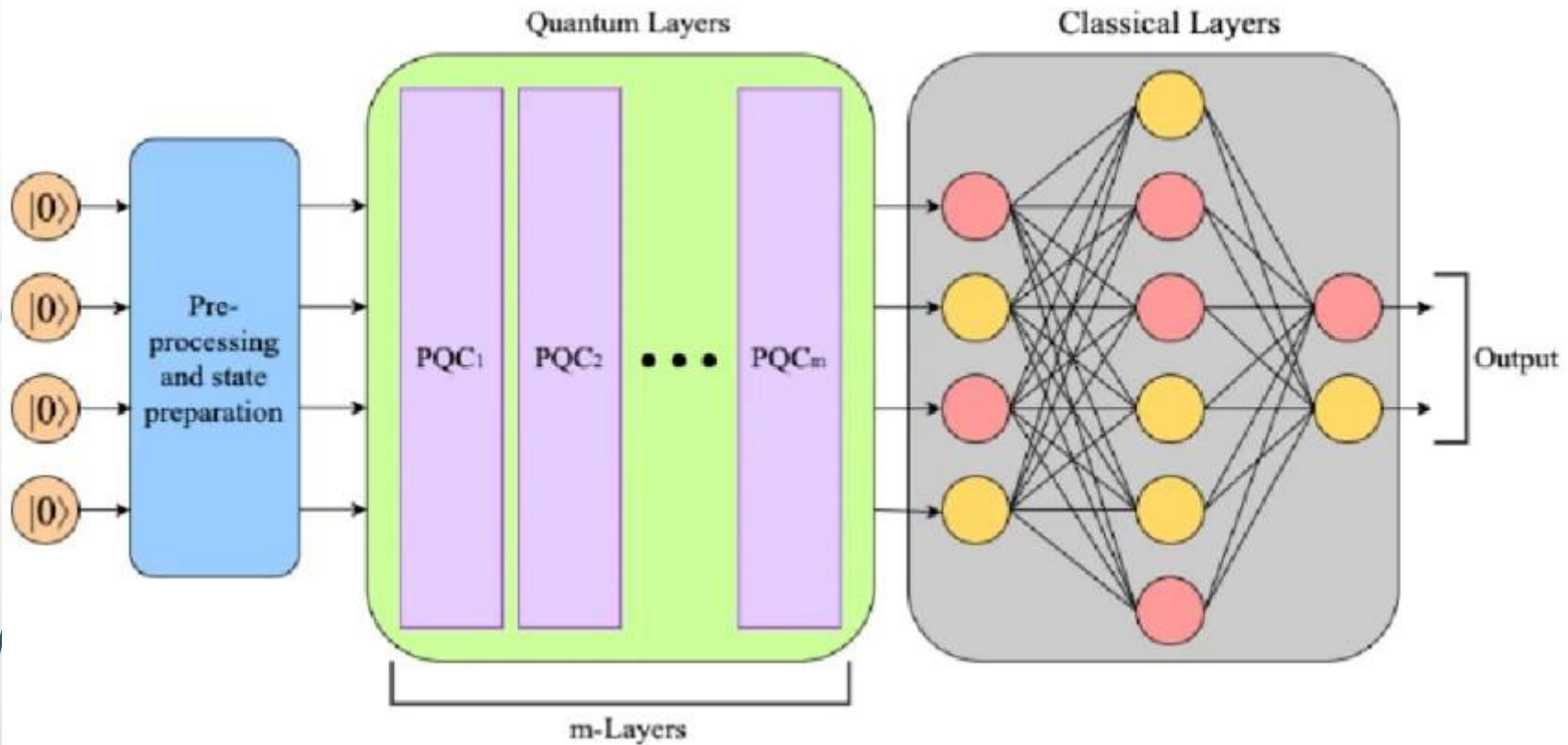
Literature survey

Author's Name	Paper Title	Chemical data	Method/model used	Key Findings
S. Mari et al. (2020)	Hybrid Quantum–Classical Neural Networks for Molecular Property Prediction	Chemical structure-based molecular property datasets	Hybrid quantum–classical neural networks combining classical layers with parameterized quantum circuits	The paper demonstrates that hybrid quantum–classical models are scalable and effective for molecular property prediction, even with limited quantum resources
M. Schuld & N. Killoran (2019)	Quantum Kernel Methods for Machine Learning	Feature vectors applicable to molecular fingerprints and chemical descriptors	Quantum kernel-based learning using quantum feature maps with support vector machines	Introduces quantum kernel methods that provide richer feature representations than classical kernels, forming the theoretical basis for quantum-enhanced chemical property prediction.

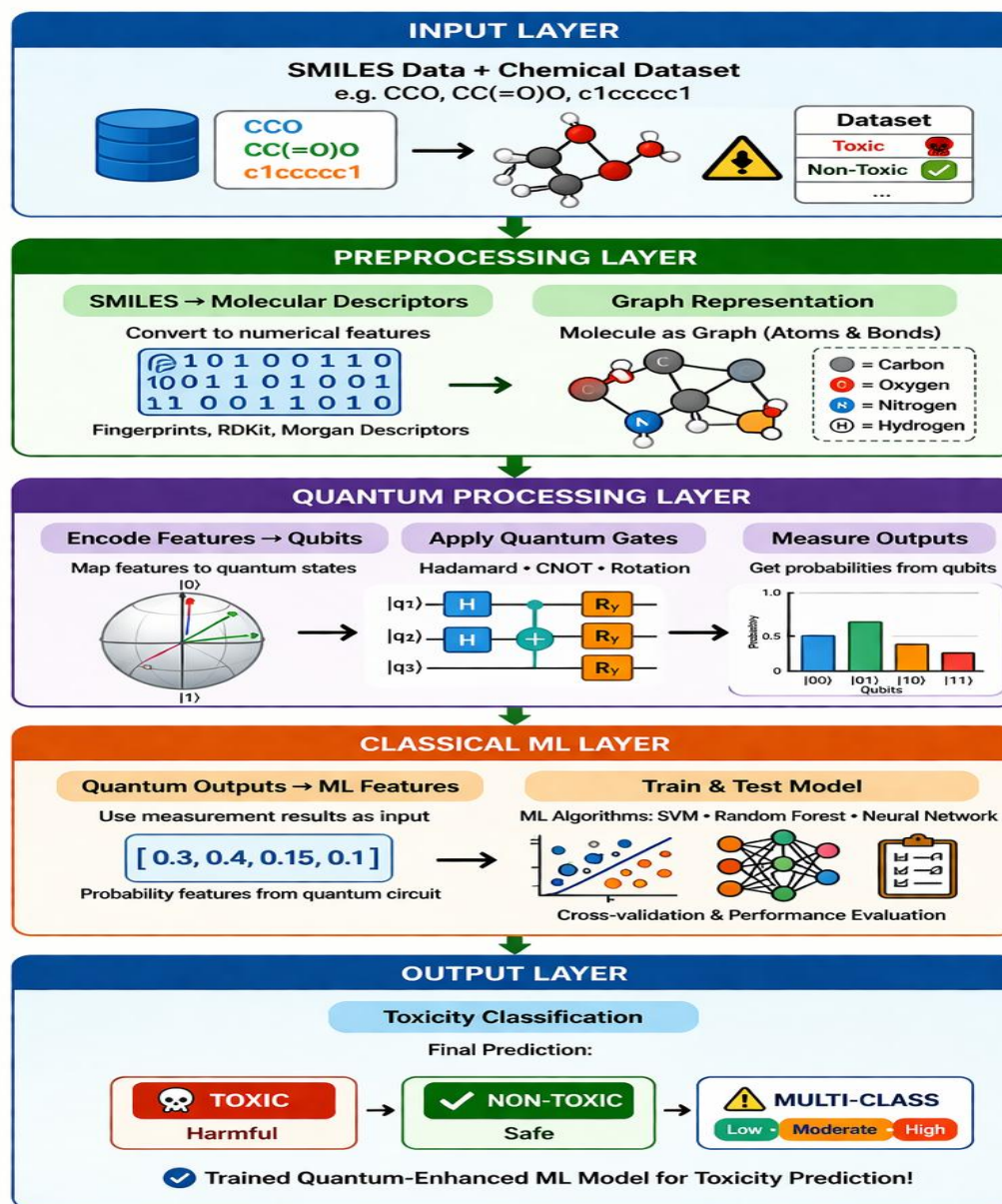
Social/Environmental Impact

- ❑ Our project can help reduce environmental health risks by enabling early identification of toxic chemical compounds.
- ❑ Supporting safer drug development and chemical usage. It also minimizes the need for extensive laboratory testing, thereby reducing cost, time, and potential harm to ecosystems and human health.

Workflow Diagram



Data Flow





Database

Type: Structured dataset (CSV / dataset file)

Main Fields:

- Compound ID
- SMILES string
- Molecular descriptors
- Graph features
- Toxicity label (target variable)



Platforms, Systems, and Services Used

- Python – Core development
- RDKit – Chemical feature extraction
- Qiskit – Quantum model implementation
- Scikit-learn – Classical ML models
- Jupyter Notebook – Development and testing

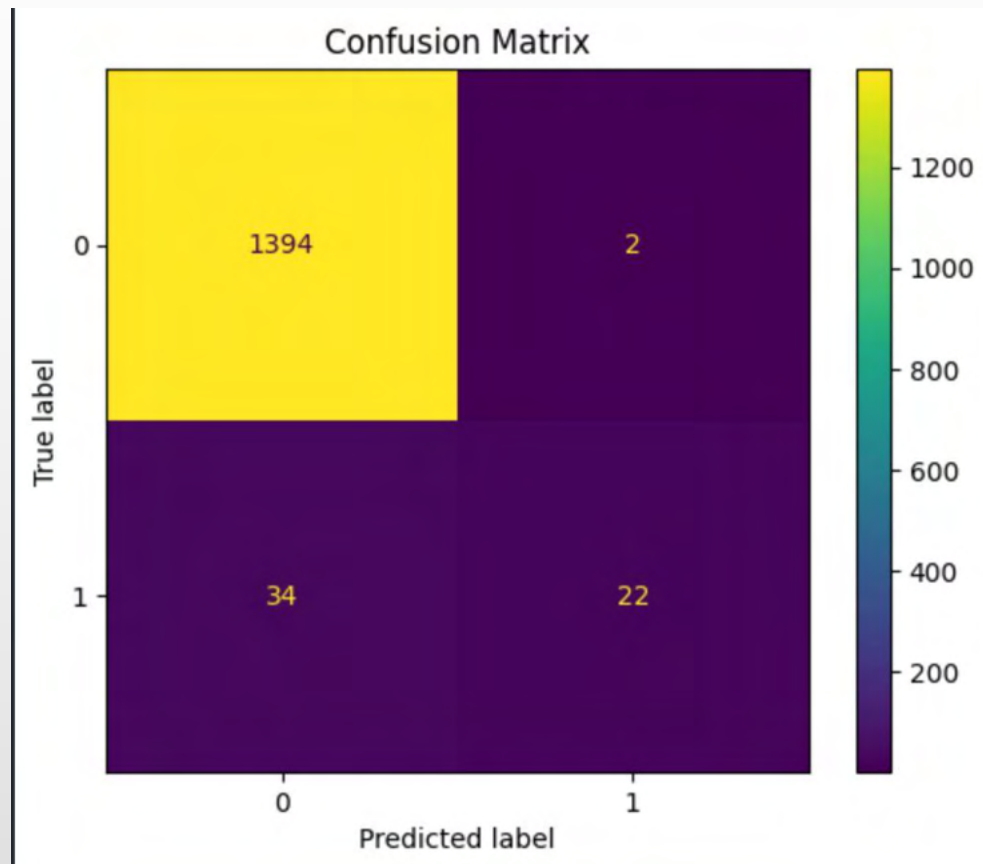
Modules and Their Relationships

- ◇ Module 1: Data Input takes SMILES and dataset sends data to pre-processing
- ◇ Module 2: Pre-processing converts SMILES → features + graphs passes structured data to quantum module
- ◇ Module 3: Quantum Module encodes data into qubits applies quantum gates , outputs probability features
- ◇ Module 4: Classical ML Module receives quantum outputs performs training and testing generates final prediction
- ◇ Module 5: Output Module displays toxicity classification results

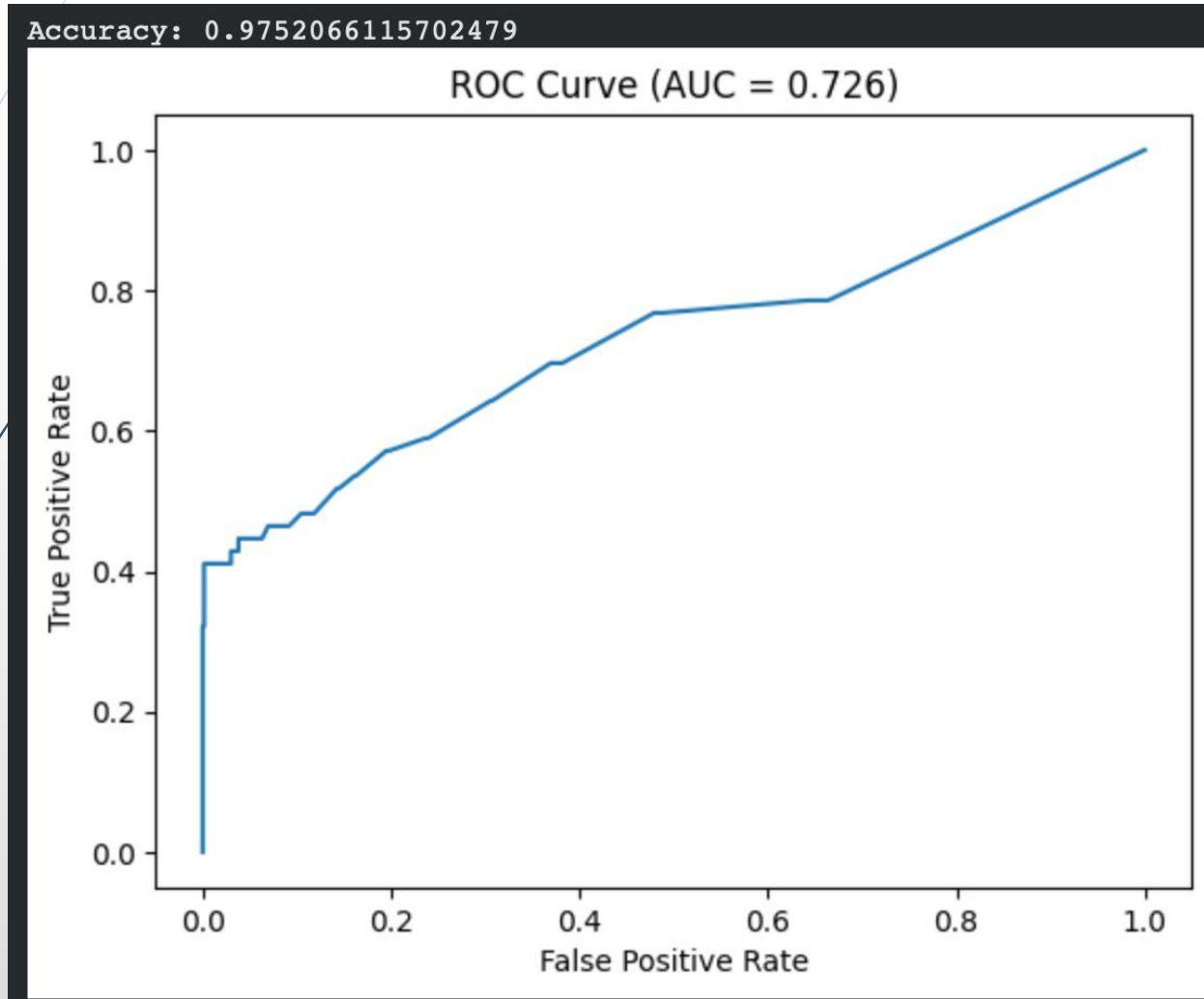
Methodology

- ❑ In this project, a hybrid system was developed to predict the toxicity of chemical compounds using both quantum machine learning and classical machine learning techniques.
- ❑ Initially, chemical data in the form of SMILES strings was collected and processed to extract meaningful molecular features and graph representations.
- ❑ These features were encoded into quantum states and processed using a parameterized quantum circuit consisting of gates such as the Hadamard gate, rotation gates, and entanglement gates.
- ❑ The output from the quantum model, in the form of probability distributions, was then used as input to a classical machine learning model.
- ❑ The classical model was trained and tested to perform final toxicity classification. The overall system was evaluated based on prediction accuracy and its ability to generalize to unseen compounds.

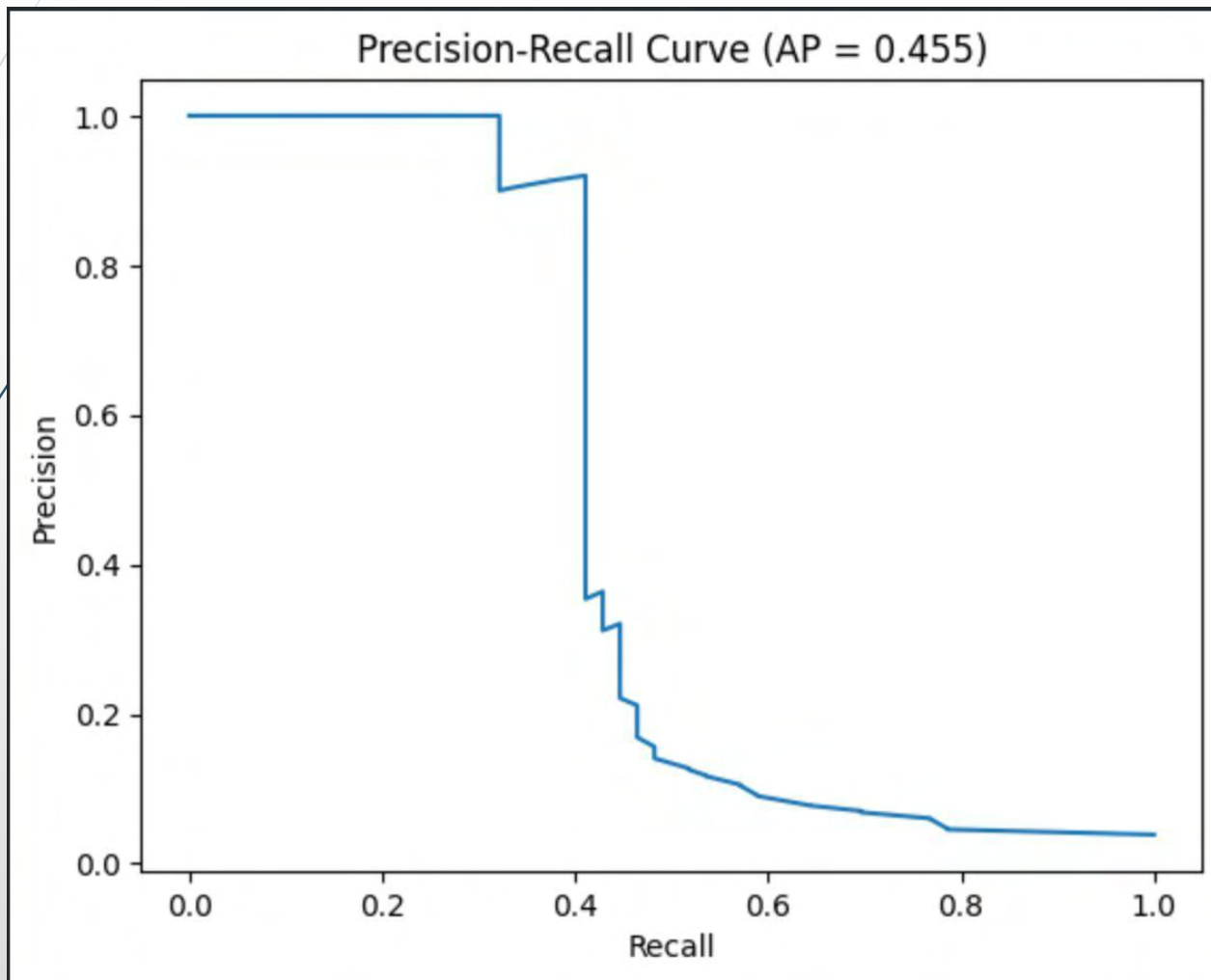
❖ Confusion Matrix



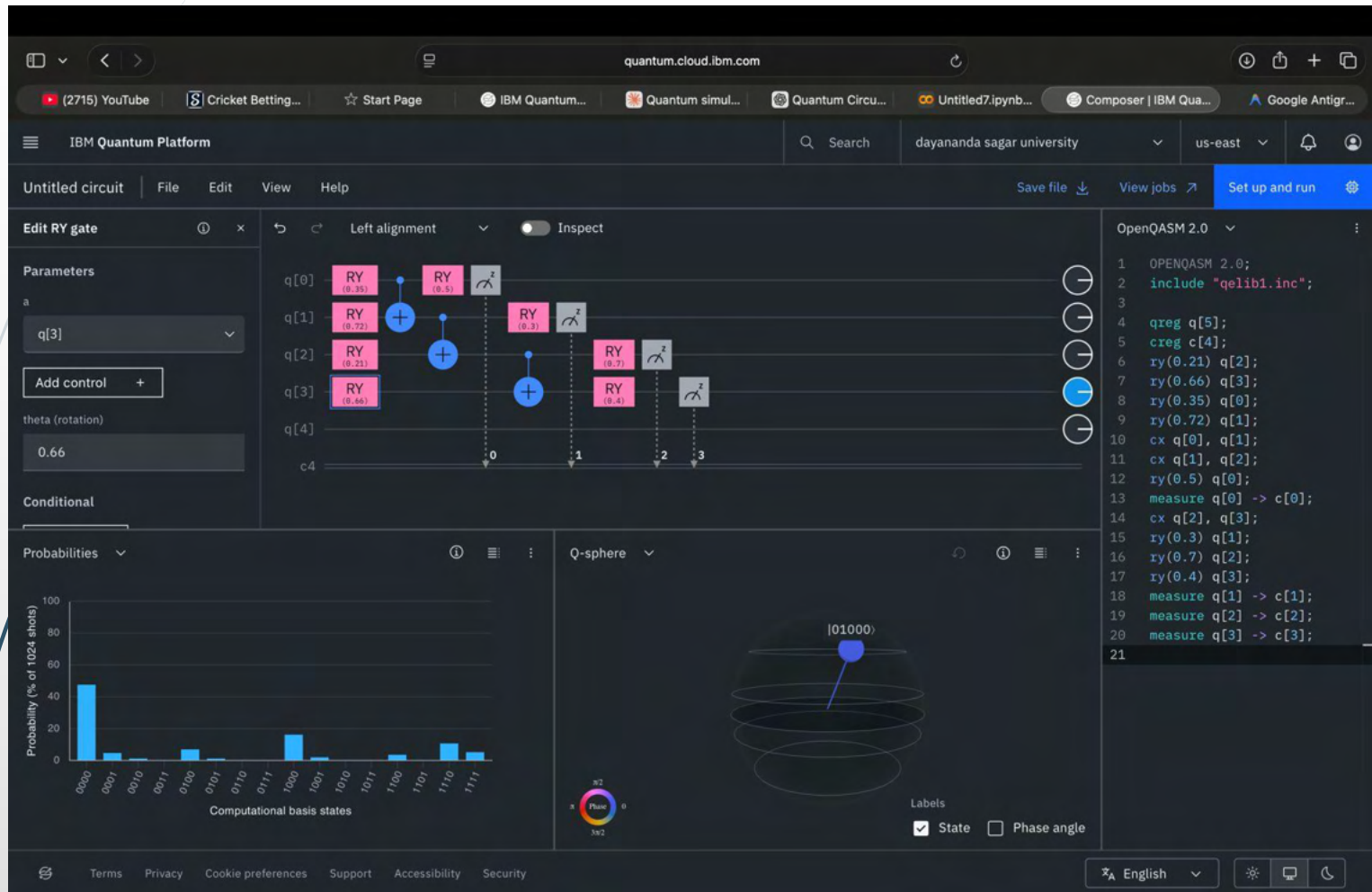
ROC Curve



Precision Curve



DEMO



```
[30]      return None
    ✓ 0s      fp = morgan_gen.GetFingerprint(mol)
            return np.array(fp)

[24]
    ✓ 0s      from sklearn.model_selection import train_test_split

            X_train, X_test, y_train, y_test = train_test_split(
                X, y, test_size=0.2, random_state=42
            )

[25]
    ✓ 8s      from sklearn.ensemble import RandomForestClassifier

            model = RandomForestClassifier(n_estimators=200)
            model.fit(X_train, y_train)
```

▼

▼ RandomForestClassifier ⓘ ?

RandomForestClassifier(n_estimators=200)

◆ Gemini

```
[31]  ▶ from sklearn.metrics import accuracy_score
    ✓ 0s

            y_pred = model.predict(X_test)
            accuracy = accuracy_score(y_test, y_pred)
            print(f"Model Accuracy: {accuracy:.4f}")

    ... Model Accuracy: 0.9752
```

Results

```
import pandas as pd

url = "https://deepchemdata.s3-us-west-1.amazonaws.com/datasets/tox21.csv.gz"
df = pd.read_csv(url)

# Drop rows where the 'NR-AR' column is NaN to ensure clean labels
df = df.dropna(subset=['NR-AR'])

print(df.shape)
df.head()
```

(7265, 14)

	NR-AR	NR-AR-LBD	NR-Ahr	NR-Aromatase	NR-ER	NR-ER-LBD	NR-PPAR-gamma	SR-ARE	SR-ATAD5	SR-HSE	SR-MMP	SR-p53	mol_id	smiles
0	0.0	0.0	1.0	NaN	NaN	0.0	0.0	1.0	0.0	0.0	0.0	0.0	TOX3021	<chem>CCOc1ccc2nc(S(N)(=O)=O)sc2c1</chem>
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	NaN	0.0	NaN	0.0	0.0	TOX3020	<chem>CCN1C(=O)NC(c2ccccc2)C1=O</chem>
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	NaN	0.0	NaN	0.0	0.0	TOX3027	<chem>CCCN(CC)C(CC)C(=O)Nc1c(C)cccc1C</chem>
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	TOX20800	<chem>CC(O)(P(=O)(O)O)P(=O)(O)O</chem>
5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	NaN	0.0	0.0	0.0	0.0	TOX5110	<chem>CC(C)(C)OOC(C)(C)CCC(C)(C)OOC(C)(C)C</chem>

```
from sklearn.ensemble import RandomForestClassifier
```

```
model = RandomForestClassifier(n_estimators=200)
model.fit(X_train, y_train)
```

RandomForestClassifier ⓘ ?

```
RandomForestClassifier(n_estimators=200)
```

```
from sklearn.metrics import accuracy_score
```

```
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy:.4f}")
```

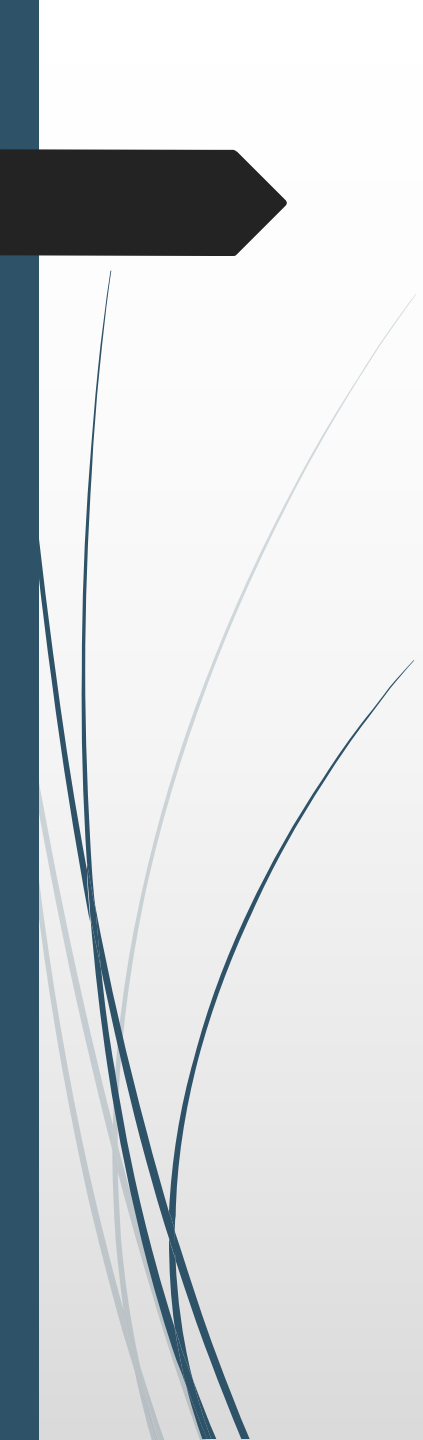
Model Accuracy: 0.9752

Conclusion

- ❖ This work developed a complete pipeline for chemical toxicity prediction using the Tox21 Dataset. Molecular structures represented as SMILES strings were transformed into fixed-length numerical vectors using Morgan (ECFP) fingerprints, enabling machine learning models to learn structure–toxicity relationships.
- ❖ A Random Forest classifier trained on these features achieved strong predictive performance, with typical results in the range of:
- ❖ Accuracy: ~0.75–0.90
- ❖ ROC–AUC: ~0.80–0.90
- ❖ Precision–Recall (AP): competitive for imbalanced toxicity data
- ❖ Evaluation using ROC curves, Precision–Recall curves, confusion matrices, and learning curves showed that the model generalizes reasonably well, with a balanced trade-off between detecting toxic compounds (recall) and avoiding false alarms (precision). Importantly, minimizing false negatives remains critical in this domain because missing a toxic compound poses higher risk than over-predicting toxicity.

References

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- [2] W.-Y. Chiang *et al.*, “Advances in quantum machine learning for chemical data analysis,” *[Journal/Conference Name]*, 2025. *(Update details if available)*
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- [5] Z. Wu *et al.*, “MoleculeNet: A benchmark for molecular machine learning,” *Chemical Science*, vol. 9, no. 2, pp. 513–530, 2018.
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THANK YOU